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An Ocean of Troubles: Advancing Marine Sociology

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ABSTRACT

We advance marine sociology to analyze the human dimensions of ocean systems. Human societies are fundamentally linked to marine systems and are transforming oceanic conditions in dramatic ways, resulting in socio-ecological problems. Despite the great and important possibilities in this realm, these relationships are seldom studied within sociology. This article highlights the ecological foundations of human societies, emphasizing interactions with marine systems, and presents marine sociology as a valuable expansion of environmental sociological studies. This approach seeks to better integrate sociological and ecological sciences. We propose that social metabolic analysis serves as one useful theoretical framework for examining the socio-ecological interrelationships. The analysis focuses on the dynamics of ocean systems, social processes that are changing marine ecosystems, and the perennial interactions within and between these systems. We provide brief analyses of climate change, ocean acidification, and pollution, revealing how the modern socioeconomic order has created ecological rifts in marine ecosystems, and how these concerns reciprocally affect social life.

KEYWORDS: environmental sociology; environmental change; socio-ecological systems; social metabolism; metabolic rift; climate change.

Marine environments have played an intimate part of human history. Yet, they remain elusive places, often seen as a realm beyond society. As recent scientific research and reports indicate, humanity's presence is far from absent in the ocean. Anthropogenic drivers, such as pollution, overfishing, and carbon dioxide emissions, have affected every area of the World Ocean. The extent of environmental change is far more extensive than previously thought, and the rate of change is faster than predicted (Halpern et al. 2008; ISPO 2013).

The ecological problems in marine systems do not begin or end at sea. They derive from social and historical processes. Further, the effects are not limited to ocean life. These changes in marine ecosystems have serious implications for terrestrial, including human, communities. Marine problems are fundamentally social problems. As the esteemed conservationist Sylvia Earle (1995) observed:

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> ocean[s] . . . provide the cornerstone of the life-support system for all creatures on our planet, from the deep-sea starfish to desert sagebrush. *That's why the ocean matters*. If the sea is sick, we'll feel it. If it dies, we die. Our future and the state of the oceans are one (p. xii, emphasis in original).

Environmental sociologists have made ecological issues a legitimate subject of social inquiry. As the field continues to develop, there has been a sizable expansion in research that addresses the human dimensions of environmental change. We contend that the extension of sociological research must continue in order to better examine the relationships between humans and oceans.

We conceive of marine sociology as a complementary and parallel development of environmental sociology. We are not proposing that marine sociology is—like environmental sociology—its own subfield. We present marine sociology as a broadening and accompanying area of study, which has not received extensive attention within the field. Thus, we provide a detailed argument regarding how we can pursue a sociologically directed study of marine systems and specific examples of analysis. There are likely many ways to accomplish this, but we offer an analytic program that provides an effective approach for marine sociological investigation.

This area of research incorporates a deeper examination of the interrelationships between social and marine systems. It recognizes that human societies are linked to ocean systems in complex ways, given direct and indirect relationships and exchanges. Also, it is noted that marine ecosystems provide important ecological services, or useful benefits that emerge from them and their processes, which support human communities.

A central aim of this article is to demonstrate the value of advancing marine sociology as it (1) provides a distinct avenue for the expansion of a socio-ecological approach already initiated, but not fully developed, in environmental sociology;[1] (2) serves as a means to illustrate the importance of socio-ecological analysis, given that marine systems and associated socio-ecological relationships, which tend to be passed over, are brought to the forefront of sociological investigation; (3) expands and encourages new research possibilities in sociology, including moving beyond existing sociological analysis of fisheries; and (4) further integrates sociology into debates and analyses on global environmental change.

In what follows, we provide a brief discussion of the relationship between marine systems and human societies. We then consider the potential challenges and barriers to a sociological analysis of oceans. By reflecting on how the rise of environmental sociology created space for important scholarship regarding ecological concerns, we argue that a similar development is necessary to direct greater attention to marine questions. Following this discussion, we contend that one useful avenue for pursuing this research is applying social metabolic analysis, as it provides a valuable theoretical approach for examining marine-society relationships. To elaborate on this, we then examine some specific problems associated with climate change, ocean acidification, and pollution that have resulted in ecological rifts in marine ecosystems. We conclude by reflecting on the significance and possibilities of marine sociology.

SOCIOLOGY, THE ENVIRONMENT, AND THE MARINE QUESTION

Humans are dependent on a vast number of ecosystem services and larger biogeochemical cycles for survival. These processes support complex social organizations. Marine systems play an important role in regulating the conditions that help sustain life on Earth. Human societies draw on marine systems for a variety of direct services, such as food, transportation, energy, and recreation. Cities line the coasts. Oceans have served as channels of transportation for seafaring civilizations and, in the modern era, they are a major part of international trade. Increasingly marine areas are supplying and being explored as sources of energy, such as fossil fuels, wind, and tidal. Oceans offer an abundance

1 We define a social-ecological approach as one that engages and incorporates ecological processes, principles, and research findings into sociology, which we elaborate throughout the article.

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of recreational activities, and numerous tourist destinations depend on clean and healthy waters for their economic welfare. Artists draw inspiration from marine landscapes and contemplate the human condition. The World Ocean has also been used as a sink for the refuse of modern production systems, such as agricultural runoff, waste from fossil-fuel production, nuclear waste, toxic pollutants, and trash in general.

Given the array of relationships, institutions, and social structures that are closely tied to and interact with marine ecosystems, and their fundamental importance for human lives, it should be evident that sociology has an important role in addressing and understanding socio-ecological issues in this realm. However, to date, social scientists have made modest contributions to these studies. Important work has been done on resource economics, and policy and cultural studies, particularly in relation to fisheries. But overall, this area of study is underdeveloped and is in need of a program that employs a socio-ecological analysis to address the larger human dimensions of oceanic change. It is worthwhile to consider the various challenges inside and outside of the discipline associated with studying marine questions, and the potential tools and avenues to pursue marine sociology.

In regard to the challenges, there are many plausible explanations. First, the ocean is commonly viewed as something far removed from human society. In some ways, it is deemed out of sight, out of mind. Terrestrial ecosystems and animals are doubtlessly considered more directly relevant to sociological matters, or even more important, as they share many physiological and geographic aspects with humans. So, to the extent that sociology incorporates nonhuman lives into social research, the analyses are inclined toward terrestrial life compared to aquatic life. Second, and related to the first point, marine systems are easily ignored as a subject of social relations. Since humans are not marine creatures, these spaces can easily be regarded as socially irrelevant. Consequently, the ocean is often seen as a realm for exclusive study by natural scientists, regardless of the findings that emphasize the importance of anthropogenic drivers of marine changes. Third, it can be difficult to bridge the social and natural sciences for both analytical and institutional reasons. Fourth, studying ocean and marine systems can be seen as theoretically and methodologically difficult for social scientists. Therefore, it is possible that part of the neglect of this topic stems from a lack of a coherent research approach.

As suggested, social science research that addresses marine concerns has primarily examined issues associated with resource management, public policy, and economic exchange (see, for example, Heal and Schlenker 2008; Hoagland, Jin, and Kite-Powell 2003; Hundloe and Arneson 2002). Much of this work is directly influenced by neoclassical economic models and/or rational choice theory, and tends to emphasize how marine fishing and aquaculture systems can be managed to expand economic investments, enhance the application of technological innovations, and conserve commercially valuable aquatic resources (such as fish species). Many of these models have a narrow conception of marine and social systems. Marine systems are largely considered instrumental resources, which are inputs into global economic supply chains where, for example, individuals rationally assess the options that are available to achieve a desired outcome in a fishery (Hundloe and Arneson 2002).

Other social science scholars have chiefly focused on political, economic, or cultural relationships, which highlight important elements that are associated with marine issues and communities, including fishing cultures, consumer concerns, public attitudes, regulatory disputes, and political processes (see, for example, Bailey 1988; Clement 2011; Constance et al. 1995; Hamilton and Safford 2014).[2] A sizeable portion of this valuable work can be loosely defined as sociology of fisheries and/or

2 Other social sciences have also developed marine studies including anthropology and human geography. Anthropology has been quite fruitful when examining the socio-structural impacts on marine systems, and has developed a subarea called maritime anthropology. This research examines fisheries systems, often from a historical or cultural standpoint (Acheson 1981; Smith 1977). Further, anthropologists and other social scientists have made enormous and important contributions to our understanding of fisheries management and concerns related to common property resources in marine zones (see McCay and Acheson 1990). Social scientists in human geography and political ecology have provided critical analyses of property regimes in the marine realm, fisheries management and governance, and aquaculture systems and ecology (Havice and Campling 2010; Jacquet 2009; Mansfield 2004, 2007).

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aquaculture (see, for example, Bailey 1988; Bush and Oosterveer 2007; Constance, Bonnano, and Heffernan 1995; Lauer 2008; Phyne and Mansilla 2003; Sinclair 1996; Vandergeest, Flaherty, and Miller 1999). This research has been crucial in laying the tracks for marine sociology.

However, a central objective of the current formulation is to promote an approach that can aid in the growth of research beyond sociology of fisheries. We contend that the few sociological studies on fishing and communities that also incorporate and address local physical conditions, ecological processes, and environmental change are pioneering the work that we suggest can be characterized as marine sociology (see, for example, Hamilton, Brown, and Rasmussen 2003; Hamilton et al. 2004; Longo 2012; Longo, Clausen, and Clark 2015). We seek to directly explicate and expand the sociological significance and analytical power of this emerging realm of study.

Natural scientists—while enormously productive regarding our knowledge of the physical world—often lack sophisticated analyses of the social drivers related to marine ecosystem change. This work tends to singly stress the importance of human population growth, or at times an ahistorical conception of human nature. For example, an essentialist explanation of human character—such as greed—is regularly used to explain overfishing, as in the "tragedy of the commons" argument (Hardin 1968). This approach is often seen as sufficient for considering social factors. Such an explanation is inadequate as far as providing a comprehensive, contextualized, and nuanced understanding of human interactions within the larger physical world. Therefore, the specific social-structural influences, institutional dynamics, and historical contexts that play critical roles in the social dimensions of environmental problems, or the social psychology of such matters, are generally overlooked.

We contend that these challenges within and outside the discipline can be effectively addressed. In fact, the historical development of environmental sociology serves as a powerful guide for how a marine sociology can be advanced. At the same time, environmental sociology provides theories and methods to forge a marine sociology, which can help bridge the sciences and increase the potential of a socio-ecological analysis of associated problems.

Environmental Sociology and the New Ecological Paradigm

In the 1970s, environmental sociologists worked to incorporate earthly questions into sociology. William Catton and Riley Dunlap (1978) argued that the twentieth-century underpinnings of social sciences were "human exemptionalist," suggesting that human society and culture in general were deemed as separate from the finite "web of nature." They suggested that social scientists tended to ignore nonhuman nature and natural limits, assuming that social processes were distinct from the biophysical world. Inherent in human exemptionalist sociology was the presumption that human ingenuity, culture, and technological innovation provided the means to overcome whatever natural constraints may arise. They proposed that this anthropocentric approach made it "difficult for most sociologists, regardless of their preferred orientation, to deal meaningfully with the social implications of ecological problems and their constraints" (Catton and Dunlap 1978:42). Given the dominant presuppositions, sociological analysis, itself, was constrained in its application and its explanations.

Dunlap and Catton (1979:251), as well as others, worked to establish environmental sociology as "the study of the interaction between society and environment."[3] They proposed that the social sciences as a whole would benefit from adopting a "new ecological paradigm" that emphasized and examined the coupling of human and natural systems (Catton and Dunlap 1980). Drawing on an interdisciplinary approach that can be broadly defined as human ecology, they clarified the analytical necessity of embedding social relations within the larger biophysical world. Humanity is physically

3 In this article, we are focused on environmental sociology rather than sociology of the environment. The former suggests that biophysical variables and relationships are an integral part of the analysis. The latter is often focused on the social and cultural relationships, and the environment is secondary. We do not disparage the latter approach to environmental scholarship, as it also offers important insights. However, we are focused here on ecological relationships, which we contend are also central to the establishment of a marine sociology.

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dependent on ecological systems, and environmental conditions establish material limits on social development. As a result, ecological relationships must be integrated into social analyses in order to offer a comprehensive assessment of socio-ecological interactions, change, and context. Their call for a radical reconceptualization of the theoretical and scientific presuppositions of social sciences contributed to a reassessment of social theory and methods. Human ecology served as a sound basis for integrating ecological elements, concepts, and principles, and contextualizing human societies in a manner that had previously been largely avoided.

While advancing the potential of socio-ecological analyses, the diffusion of a new ecological paradigm has been slow and limited in sociology. Environmental sociology research has unquestionably created more space for the examination of socio-ecological systems and relations, breaking down previous barriers, expanding the realm of social inquiry, and opening up the possibility for new theories and approaches. Human ecologists led the charge. Since then, distinct theoretical traditions have been established.

Treadmill of production scholars broadly examine how the growth imperative of capital escalates the demands for resources and expands the amount of pollution, resulting in general patterns of environmental degradation (Gould, Pellow, and Schnaiberg 2004:297). Ecological modernization researchers study "how various institutions and social actors attempt to integrate environmental concerns into their everyday functioning, development, and relationships with others, including their relation with the natural world" (Mol, Spaargaren, and Sonnenfeld 2009:4). Social metabolic scholars share with treadmill of production theorists a critique of political economy, but further emphasize the flows and exchanges of matter and energy. Scholars working within established research traditions, such as world-systems analysis and world society, have analyzed how global relations, such as trade and treaties, influence environmental outcomes (Grimes and Kentor 2003; Meyer et al. 1997; Roberts and Parks 2007). Environmental justice scholars have emphasized the disproportionate impacts of environmental problems associated with race, class, and gender (Mohai, Pellow, and Roberts 2009).

In engaging these theories, environmental sociologists have examined a variety of environmental concerns such as the social drivers of environmental change; the role of international organizations, domestic institutions, and citizens in facilitating/hindering the expansion of environmental policies; and social inequalities associated with environmental hazards. As a result, many important research studies in environmental sociology have been published in leading sociology journals (e.g., Jorgenson and Clark 2012; McCright and Dunlap 2000; York, Rosa, and Dietz 2003). Nevertheless, the development and shift to a new ecological paradigm is still in progress. This step requires a theoretical apparatus that ecologically deepens the analytical potential of sociological studies, merging the material basis of human existence together with the social processes associated with the larger-scale historical structures that influence the interpenetration of society and nature—or what we call a socio-ecological approach.

Marine sociology further develops the new ecological paradigm and environmental sociology, as far as conceiving of oceans and marine systems as sociological subjects for inquiry. It addresses some enduring limitations of sociological analyses, such as persistent remnants of human exemptionalism. While multiple theoretical and research traditions could and should be engaged to study marine systems, here we apply a social metabolic analysis to illustrate an environmental sociological approach that can lead to innovative scholarship on oceanic systems. This perspective can serve as a constructive foundation for marine sociology, on which others may elaborate. Therefore, we provide a discussion of the background and selected application of social metabolic analysis to demonstrate this point.

Social Metabolic Analysis

Like human ecology, social metabolic analysis embeds human society within the larger biophysical world and studies the interpenetration of social and ecological systems. The approach emphasizes that ecosystems and natural cycles operate independently and in relation to human society and other systems. Social metabolic theorists incorporate an analysis of these cycles and processes, noting the

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regulatory dynamics that contribute to the regeneration and conditions of ecosystems. These theorists also highlight how political-economic systems mediate human interactions and exchanges with the larger world (Clark and York 2005; Foster 1999). Given this general orientation, Marina Fischer-Kowalski (1998:62) proposes that metabolic analysis serves as an effective approach for investigating the coupling of human and natural systems and can "cut across the 'great divide' between the natural sciences . . . and the social sciences," an important matter for achieving our suggested goals with the advancement of marine sociology.

In the 1800s, physiologists employed the concept of metabolism to examine the biochemical processes between a cell and its surroundings, as well as the interpenetration and exchanges between an organism and the biophysical world. Karl Marx, who closely followed scientific debates and discoveries, incorporated the concept of metabolism into his critique of political economy, explaining that he "used the word in the 'natural' process of production for the exchange of matter between man and nature" (Marx and Engels 1975, vol. 24:553). He was clear that humans are dependent on nature and "can create nothing without" it (Marx 1964:109). As a result, there is a necessary "metabolic interaction" between humans and the earth. Labor serves as "a process between man and nature, a process by which man, through his own actions, mediates, regulates and controls the metabolism between himself and nature" (Marx 1976:283). The labor process, including the exchange with ecological systems, is influenced by the dominant socioeconomic arrangements and social institutions, giving a distinctive character to what Marx saw as the social metabolism.

The nuance of Marx's conception of metabolic relations is evident in his conception of both a "universal metabolism of nature" and a social metabolism. He noted that there is a "universal metabolism of nature" within the broader biophysical world (Marx and Engels 1975, vol. 30:54-66). Specific cycles and processes operate that constitute, maintain, and may regenerate ecological conditions. Human society exists within the earthly metabolism and continually interacts with it. Thus, the social metabolism of human systems takes place in relation to the universal metabolism of nature, which in part establishes ecological limits. Marx avoided subsuming society into nature, as well as vice versa, in order to evade "the pitfalls of both absolute idealism and mechanistic science" (Foster 2013:8). Therefore, while providing analytical distinction, the approach recognizes that humans and the rest of nature are in constant interaction, resulting in reciprocal influences, consequences, and dependencies. These processes emerge within a relational whole, or the universal metabolism of nature.

István Mészáros (1995) and John Bellamy Foster (2000) explain that each mode of production generates a distinct social metabolic order that influences the interchange and interpenetration of social and ecological systems. These theorists emphasize that capitalism, as the dominant socioeconomic system today, tends to direct socio-ecological interactions and exchanges toward capital expansion and accumulation. To illustrate some of the metabolic changes that accompanied the rise of capitalist social relations, it is useful to consider how Marx—drawing on the work of agricultural chemists and agronomists—analyzed transformations in agricultural production (Foster 2000; Foster, Clark, and York 2010). As Marx (1971) recognized, soil "fertility is not so natural a quality as might be thought, it is closely bound up with the social relations of the time" (pp. 162-63). For example, in many societies, farm animals were directly incorporated into agricultural production. They were fed grains from the farm and their manure, which contained nutrients, was reincorporated into the soil as fertilizer.

This particular metabolic interchange was reconfigured in large part by the enclosure movement, the rise of the new industrial and institutional systems, and social relations associated with capitalist development. Marx (1976) explained that these changes also magnified the division between town and country. Food and fiber from farms were increasingly shipped to distant markets, transferring the nutrients of the soil from the country to a distant city. Much of the nutrients were squandered, as waste was not returned to the soil. Instead, waste accumulated as pollution within cities and waterways (Foster 2000; Mårald 2002). New agricultural practices, including the application of industrial

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power, increased the scale of operations, transforming and intensifying the social metabolism while exacerbating the depletion of the soil nutrients (Marx 1976).

Marx (1976) explained that capitalist agriculture progressively "disturbs the metabolic interaction between man and the earth" (pp. 637-38). In other words, capitalist agriculture and the various mechanisms used to intensify production and increase profits created a "metabolic rift" in the soil nutrient cycle, diminishing the nutrients that supported the production of crops (Foster 1999). As it violated the universal metabolism associated with the soil nutrient cycle (also conceived as a law of restitution), it undermined soil fertility and the conditions that supported human society.

Recent metabolic research has extended the analysis to address transformations in climate, hydraulic, and forest systems (Austin and Clark 2012; Clark and York 2005). Much of this work examines how the social metabolism of capital has created specific environmental problems, as it transgresses against the universal metabolism of nature. The intensification of the social metabolism demands more energy and raw materials, generating an array of ecological contradictions and ecological rifts (Burkett 2006; Foster et al. 2010).

The social metabolic approach can serve as a basis to better extend social analysis to marine systems and illuminate important dimensions of this socio-ecological relationship. In applying this approach, human society is examined as part of a larger ecological complex, which consists of natural cycles and processes that operate independently and in relation to social developments. The interchange of matter and energy between social formations and the larger environment is emphasized. The framework accounts for historical transformations in socio-ecological systems and deep structures—whether in nature or society—that exert influence over long periods of time. It considers complex interactions, identifies direct and indirect relationships, and recognizes the importance of feedback loops within and between the social and physical. Given the importance of marine systems for social development and the significance of anthropogenic forces transforming ocean conditions, it is a realm that is ripe for this manner of sociological investigation.

AN OCEAN OF RIFTS

In the modern era, marine systems are being transformed in a manner unmatched by any previous time in human history. The scale, scope, and pace of these changes are remarkable. Numerous scientific research findings signify the extent of human impacts and the importance of social analysis on the marine front. The opportunities for social research are wide-ranging. In what follows, we briefly discuss a few of the areas of concern in marine systems that are linked to social processes. We simply highlight a small number of these issues associated with climate change, ocean acidification, and pollution, and the human dimensions of these concerns, as a way to promote more exhaustive studies in marine sociology.[4] We incorporate social metabolic analysis as a thread, demonstrating one useful approach for advancing marine sociology.

Climate Change and Ocean Acidification

Anthropogenic forces, such as the burning of fossil fuels and deforestation, are increasing the concentration of carbon dioxide and other greenhouse gases in the atmosphere, causing global climate change (Stocker and Qin 2013). Since the Industrial Revolution, humans have been using fossil fuels as a means of expanding systems of production, consumption, and transportation. As a result, "atmospheric carbon dioxide (CO_2) has increased from 278 to over 400 parts per million (ppm) over the industrial period" (Gattuso et al. 2015). Environmental sociologists have determined that underlying social forces influence demographic trends, the scale and structure of economic operations (including

4 In previous research, we discussed how capitalist expansion and intensification of fishing operations contributed to technological innovations that allowed fishers to capture massive quantities of marine life (Longo et al. 2015). Many fisheries are stressed, given that fish are being harvested faster than they (and before they can, in some cases) reproduce. Thus, overfishing creates a distinct ocean rift, which can lead to fisheries collapse.

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production, trade, and consumption), and technological innovation and deployment, which in turn act as drivers of global climate change (Rosa et al. 2015). Metabolic scholars highlight how the growth imperative of capitalism, which shapes the social metabolism, has broken the solar-income budget—extracting carbon that had been buried and releasing it into the atmosphere at an unprecedented rate—and created a metabolic rift in the carbon cycle (Clark and York 2005). Ocean systems play a significant role in climate regulation and climate change. Climate research has indicated that the other major carbon dioxide problem is ocean acidification, which presents a major threat to Earth systems. The dynamics of the social metabolic order of capital play a central role in both of these socio-ecological problems.

"For the globe as a whole," wrote Rachel Carson (1991:170), "the ocean is the great regulator, the great stabilizer of temperatures." The oceans influence the concentration of carbon dioxide in the atmosphere as the gas is continuously exchanged at the surface. The World Ocean absorbs, through the air-sea interchange, from a third to half of the carbon dioxide produced. However, marine temperatures influence the amount of carbon dioxide that the oceans can absorb. As ocean temperature increases due to climate change, the absorptive capacity of ocean systems decline. A feedback loop is in operation, which is affected by human social action. A report from the Royal Society clearly states the consequences: "A decrease in the amount of CO_2 absorbed by the oceans will mean that relatively more CO_2 will stay in the atmosphere. This will make global efforts to reduce atmospheric concentrations of CO_2 and the associated climate change more difficult" (Raven et al. 2005). Feedback loops, such as the example just described, or the thawing of permafrost releasing more greenhouse gases, and the melting of ice sheets and glaciers reducing reflectivity of Earth's surface (the albedo effect), can intensify climate change. As human activities cause the concentration of carbon dioxide in the atmosphere to increase, irreversible transformations are happening in the Earth system. The social metabolic order of capital has resulted in interactions with these systems that drive unprecedented marine ecological change.

Climatic changes, including changing water temperatures, are leading to the melting of ice sheets, including Greenland and Antarctica (Joughin, Smith, and Medley 2014; Rignot et al. 2011). The rate of melting of the West Antarctic ice sheet indicates the onset of collapse. Recent modeling suggests that climate forcing and associated affects on ocean currents in the North Atlantic and Southern Ocean will increase both severe storms and ice sheet melting, which can result in "non-linearly growing sea level rise, reaching several meters over a timescale of 50-150 years" (Hansen et al. 2016:3762). It is becoming ever-more certain that the melting of major ice sheets will contribute to a significant rise of sea level over the next century and forced migrations of humans along coastlines, especially in parts of the world where political-economic conditions limit potential response strategies. These changes increase the social and ecological vulnerability of coastal zones already ecologically transformed by modern economic development. This has lead to attempts at technological fixes, such as carbon sequestration, geo-engineering, or sea walls, which ignore the root socio-ecological problems as they are regarded as irreconcilable with the growth imperative of capital. All the while, carbon dioxide emissions continue to grow. It is clear that sea-level rise is a serious concern for coastal communities and major cities throughout the world. Sociologists can be at the forefront of understanding and addressing these oncoming challenges by better analyzing the socio-ecological processes that drive them, and also constrain effective responses.

Another concern is that warming oceans will hold less oxygen, which may promote and sustain hypoxic conditions associated with other human-induced conditions, such as nutrient runoff. A shoaling effect reduces available habitat for pelagic fish species, such as tuna and marlin, and forces them closer to the surface in search of oxygen rich habitat. Consequently, these species become more vulnerable to overfishing (Stramma et al. 2012). The expansion of hypoxic conditions associated with warming temperatures is projected to have enormous consequences for fisheries. Vast areas of oceans are likely to become inhospitable to the marine life that humans draw upon for food. These changes

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will certainly have consequential social outcomes, particularly for fishing communities. As the changes take affect, sociologists must increasingly include these ecological factors in their analyses.

The accumulation of carbon dioxide in the atmosphere is contributing to changes in ocean chemistry. As carbon dioxide reacts with water, it results in an increase in hydrogen ion concentration (H^+). As this happens, the pH of water is lowered and the concentration of carbonate ions is reduced (CO_3^{2-}) (Guinotte and Fabry 2008). This process is known as ocean acidification, and has been called "the other CO_2 problem" (Doney et al. 2009). It is a consequence of the same anthropogenic activities that are causing climate change (Raven et al. 2005). As carbon dioxide is increasingly added to the atmosphere from human activities due to the social metabolism of the dominant socioeconomic system, which includes ceaseless economic growth fueled by the burning of fossil fuels, it also concentrates in the oceans. Only recently has ocean acidification started to receive increased attention in the natural sciences and to some degree the popular media. However, social scientists have not addressed this issue yet in the growing literature on the human dimensions of environmental change, despite the far-reaching consequences.

Johan Rockström and his colleagues (2009) indicate that ocean acidification, like climate change, is one of nine measures of "planetary boundaries." These are crucial indicators for understanding the potential for maintaining an earth-system environment in which humanity can exist "safely." The current rates of change in the chemical makeup of marine systems are "unprecedented" in human history (Guinotte and Fabry 2008:322). For approximately the last 800,000 years, the acidity of the surface ocean has been "remarkably stable" (Turley and Gattuso 2012). Over the last two to three centuries, the ocean pH has decreased by .1, which is about a 30 percent increase in hydrogen ion concentrations since preindustrial levels (Guinotte and Fabry 2008). Because pH is measured logarithmically, one pH unit represents a ten-fold difference. Scientific models predict that this trend will continue, with another potential drop of .3 to .4 pH units over the next several decades (Feely, Doney, and Cooley 2009). Under the social metabolic order of capital, the upper ocean may experience "a 150 percent increase in acidity compared with preindustrial times," resulting in serious effects on marine life (Hardt and Safina 2010:69).

Ocean acidification has dramatic effects on marine calcifiers—shell and plate forming marine organisms, including mollusks, plankton, echinoderms (e.g., sea urchins), and corals, that draw on calcium carbonate ($CaCO_3$) in sea water to build their shells and plates. Lower carbonate ion concentration leads to a reduction in the calcium carbonate saturation state. As a result, calcifiers must use more energy in order to produce biogenic calcium for shell and plate formation. As specific oceanic conditions change, these organisms are increasingly susceptible to stressful conditions (Bednaršek et al. 2014). It is important to note that many of the species affected are at the base of the marine food web, and crashes to these populations could have wide-ranging ecosystem impacts, which would not remain contained to marine systems.

The social metabolism of the capital system is profoundly disrupting the universal metabolism associated with the oceans, affecting the conditions that allowed for the flourishing of marine life. While the exact ecosystem effects of ocean acidification are largely unknown due to the unprecedented and relatively recent nature of this development, research is indicating that these changes could have enormous impacts on the species that inhabit marine ecosystems (Caldeira and Wickett 2003). In fact, ocean acidification is regarded as a major factor that contributed to previous mass extinctions, and is seen as an important part of the ongoing sixth extinction (Kolbert 2014).[5]

For example, coral reef ecosystems—already under pressure from overfishing, pollution, and increasing ocean temperatures that contribute to coral bleaching—will be further stressed by ocean acidification (Gattuso et al. 2015; Hoegh-Guldberg et al. 2007). Coral ecosystems play an important

5 The ongoing sixth extinction is a term used to describe the process of species loss occurring in the modern era. Species extinction is currently occurring at a rate that is 100 to 1,000 times the background rate of extinction, or the rate of extinction that has been revealed in the fossil record.

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role in maintaining marine biodiversity, supporting an extensive web of ocean life, generating important nutrients, and providing protection to coastal communities and ecosystems from, for example, storms that are expected to increase in power in a warming world. Ocean acidification hinders the building of coral reefs. The cascading ecological rifts are clear: when coral reefs collapse, millions of other species that are dependent upon them also die (Edinger et al. 1998; Folke et al. 2004; Hallock 2005). Human coastal communities, particularly in the Global South, are affected by the impacts on ecosystems and the services that they provide, including declining fisheries captures (Gattuso et al. 2015).

Thus, an expected consequence of the changing ocean chemistry and temperature is decreasing biodiversity. For example, some scientists have argued that ocean acidification has contributed to the increasing frequency and duration of jellyfish blooms, as these species can occupy ecological space created by the loss of other species that are directly and indirectly valuable to humans (Atrill, Wright, and Edwards 2007; Gershwin 2013). Planktonic life that is the basis of fisheries food webs are negatively affected; however, the changing ecological conditions produced by anthropogenic forces open up room for jellyfish, which are predators of zooplankton, fish eggs, and larvae. This dynamic promotes the reordering of the ecological system in a manner that decreases biodiversity and is less favorable to ecosystem conditions we have come to expect in many locations, and likely the well-being of human social systems (Guinotte and Fabry 2008).

Together with the broader pressure on fish stocks from overfishing, fisheries that have been major sources of food, economic activity, and culture, such as the Atlantic cod or walleye pollock in the Bearing Sea, are increasingly less likely to supply the benefits that they have in the past (Gershwin 2013). For example, warming oceans have been linked with inhibiting the return of the Atlantic cod in the Gulf of Maine (Pershing et al. 2015). Multiple forms of socio-ecological effects are intersecting in marine systems, emphasizing the complexity of these relations and the need for multidimensional and integrative analyses.

Changes in carbon dioxide levels and ocean acidification will have several affects on fish species (including behavioral, neurosensory, and metabolic rate, among others), which will have unknown consequences for short- and long-term survival. As suggested above, the effects on shell-forming organisms, such as oysters, clams, crabs, and lobster, will have more immediate consequences. In the social metabolic order of capital, aquaculture systems have been seen as a solution to fisheries depletion. Yet, these techno-fixes are plagued with their own set of socio-ecological contradictions (Longo et al. 2015).

Oyster populations in the Pacific Northwest have experienced die-offs that are linked to ocean acidification. Hatcheries in Washington and Oregon have been stressed by the upwelling ocean systems of the region, which bring more acidic conditions to these waters. As a result, some oyster farmers are being forced out of the area, and a multimillion-dollar industry is threatened (Welsh 2013). These outcomes provide what may be the first glimpse into the social consequences of marine transformations related to ocean acidification. The changes taking place in marine ecosystems, due to climate change and ocean acidification, "increase the potential for an overall simplification of ecosystem structure and function with reduced energy flow among trophic levels and little scope for species to acclimate. The future simplification of our oceans has profound consequences for our current way of life, particularly for coastal populations and those that rely on oceans for food and trade" (Nagelkerken and Connell 2015).

Climate change and ocean acidification are directly connected to anthropogenic drivers associated with the social metabolic order of capital. The social metabolism is disrupting the universal metabolism that has long supported the conditions of marine life and coastal communities. As the chemistry of the atmosphere and the ocean changes, vast ecosystems that contributed to the rich biodiversity of the oceans are critically altered. Human livelihoods and cultures that are dependent upon marine systems are already affected by ecological problems associated with climate change and ocean acidification (Kerr 2010). As sociologists, we must expand our analyses to include these marine processes

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and better address serious socio-ecological problems. By developing a socio-ecological approach within marine sociology, the drivers and consequences of these serious concerns become apparent. Proposed social solutions and policies will need to better integrate marine ecological data into the analyses.

Marine Pollution

The World Ocean has long been intentionally and unintentionally used as a sink for waste. Human sewage has been directed to rivers and flowed to oceans, and nutrients have been dispersed into the vast marine waters. Radioactive/nuclear waste was placed in containers and disposed of in oceans. Carson protested this dumping, challenging the assumption that such actions did not have repercussions. She highlighted how radioactive materials traveled through species within ecosystems, including humans (Carson 1991:xii). Oil exploration and transportation has led to long-term pollution, ecological degradation, and the deaths of marine species, particularly following accidents such as the Exxon Valdez oil-tanker spill and the Deepwater Horizon oil-drilling rig explosion.

Marine pollution is quite sociologically revealing. It arises directly and indirectly from anthropogenic sources. Oftentimes it involves waste from terrestrial activities that eventually ends up in the oceans. Barry Commoner (1971:33) argued that "everything is connected to everything else"—this is the first of his four informal laws of ecology. Following this law, he suggested that it is necessary to study the interconnections within and between ecosystems. Social processes usually have multidimensional impacts on ecosystems, which then return to affect social processes. The list of pollution that enters the World Ocean is long. Here we highlight two persistent forms of pollution—nutrient runoff and plastic—to address specific ecological rifts in ocean systems.

In addressing marine pollution, direct connections can be made in regard to how the social metabolism disrupts the universal metabolism associated with various ocean ecosystems. The metabolic rift in the soil nutrient cycle that arose with capitalist agricultural development has not been mended, but transformed and extended. Synthetic fertilizers are used to enrich nutrient poor soils to maintain food production (Magdoff 2011; Mancus 2007). The heavy application and overuse of fertilizers on farmland has led to the runoff of nitrogen and phosphorous into streams and rivers, and eventually marine waters (Manuel 2014). Nutrient pollution of waterways and oceans causes eutrophication, which is one of the leading causes of ecological disruption in marine ecosystems. The fertilizers spur the massive growth of plant-based organisms, such as algae. As large concentrations of algae die and decompose, oxygen levels in the surrounding waters are lowered, and combined with other ecosystem stressors create hypoxic conditions.

Under typical ecosystem conditions such events are unusual. But the social metabolic processes and exchanges of the current socioeconomic order have made these events common and widespread. Nutrient pollution has been driven by the mass production of agricultural products in a manner that has paralleled factory production systems. Monocropping, concentrated animal feeding operations, and industrial-scale technologies and practices have created the conditions that lead to the heavy use of synthetic fertilizers and the collection of immense pools of animal waste, which contribute to contamination of waterways and oceans. Thus, these algal blooms are the result of unsustainable practices in food production, such as those that are employed in the "Corn Belt" of the United States. They create areas within the ocean that are inhospitable to most marine species. "Dead zones" kill marine life and can produce biotoxins. Since the 1960s, these dead zones have "spread exponentially," and over 400 dead zones have been identified around the world. In 2008, it was estimated that over 245,000 square kilometers of dead zones are affecting marine ecosystems (Diaz and Rosenberg 2008).

Harmful algal blooms are also increasing, escalating the risks of human exposure to dangerous toxins through consuming seafood, direct contact with marine water, and aerosol inhalation from wave action (Backer and McGillicuddy 2006). These so-called red tides produce toxins that have affected

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coastal communities and sea life. For example, in the summer of 2015, a large algal bloom in the Pacific Northwest, ranging throughout the region from Northern California to Alaska, prompted the Washington Department of Fish and Wildlife to shut down crab fishing and clamming in all waters, and the California Department of Health issued an advisory against consuming shellfish and small fish, like sardines (Ferris 2015). Similar phenomena have been implicated in the death of sea life, including sea mammals and large fish kills in the Gulf of Mexico, and can result in human health advisories associated with clusters of respiratory illnesses due to exposure to these toxins (CDC 2008; Flewelling et al. 2005).

Social processes on land are altering marine processes, creating conditions that repeatedly stress marine life. These changes affect human society, as employment, food, and health are harmed. The metabolic rift in the soil nutrient cycle and this ocean rift are historically linked to specific food production practices that disrupt the universal metabolism of nature. The social and ecological consequences are significant.

The second informal law of ecology is "everything must go somewhere" (Commoner 1971:39). Plastic is a product of petrochemical production processes, and has been a key component in modern production practices. It does not readily break down within a meaningful human timeframe. Thus, it remains ever present, and has become a universal part of the World Ocean. Plastic pollution presents a real challenge that is contributing to another distinct ocean rift as it enters the food chain.

In the oceans, high-pressure zones, known as gyres, help concentrate the plastic that is adrift into giant "garbage patches" (Moore 2011). The garbage patch in the Pacific Ocean is estimated to be twice the size of Texas and can reach thicknesses of 60 to 90 feet. Both large and small plastic particles populate the sea. Nevertheless, a synthetic soup is being created, as plastics are fragmented into tiny pieces. The plastic contains and absorbs chemicals—such as persistent organic pollutants—making them "supersaturated poison pills" (Casey 2007:16). Marine animals eat this plastic, perceiving it as a source of food. The chemicals accumulate through the food web and can act as endocrine disrupters. Wildlife consumes numerous pieces of plastic, resulting in the death of "more than a million seabirds, 100,000 marine mammals, and countless fish . . . in the North Pacific each year" (Casey 2007:11-12). In addition to harming marine populations, there is a danger to humans who consume seafood, given the bioaccumulation process.[6] Serious concerns are being raised regarding how the chemicals act on human bodies, as they interact with hormones, damage chromosomes, and foster mutations (Steingraber 2010).

There is no effective clean up method for plastics in the ocean. Every wave washes up more bits of plastic, especially on the islands of the South Pacific. The petrochemical industry has created a huge range of plastic material, which has been actively incorporated into consumer goods throughout all of society. In the modern social metabolic order, plastic is a throwaway product that accumulates within the global ecosystem once it is produced, altering the universal metabolism of ocean systems. Today over 72 billion plastic bottles and 500 billion plastic bags are produced each year (Casey 2007; Eriksen et al. 2014; Jambeck et al. 2015; Moore 2011). These are just the tip of the plastic iceberg. Even with the growth of plastic recycling, the ongoing production of plastic—driven by the social metabolism of capital, where for example cheaper plastics substitute for other less toxic, but more costly, products—persistently contributes to the worsening of this particular problem. In just a few decades, since the widespread manufacture and use of plastics, a colossal ocean rift has been created that is damaging biological systems, disrupting reproduction, and disturbing the life cycles of marine organisms and humans.

6 Bioaccumulation is the process by which toxics build up in an organism, as they are absorbed at a rate faster than they are lost. In addition, toxics that enter at lower levels of the food web accumulate as they move up into organisms higher in the tophic levels. Given this process, predators and other higher trophic-level species are more likely to contain elevated levels of toxic substances. This aspect of the bioaccumulation process is often used interchangeably with the term biomagnifcation.

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CONCLUSION: TOWARDS A MARINE SOCIOLOGY

Marine ecosystems are intimately linked to human societies. Unfortunately, sociologists have largely avoided studying marine systems in a manner consistent with the development of what Catton and Dunalp (1980) called a new ecological paradigm. We argue that it is important to further extend social analysis to the oceans, and develop a critical approach we call marine sociology as a way of advancing environmental sociology as an integral part of comprehending and addressing complex ecological concerns. Such an analysis cannot be left to the natural sciences alone. Sociologists need to be more active in this research and these discussions. This move is part of overcoming the bias against addressing and including biophysical relationships and processes in social investigations, originally presented by some of the principal pioneers in environmental sociology. Elaborating on marine sociology is an effort to (unapologetically) push sociological investigation into deeper ecological terrain, and to contribute to the further development of a new ecological paradigm. It promotes the expansion of environmental sociology and serves as a way to link sustainability sciences and sociological studies of coupled human-natural systems (Liu et al. 2007).

Marine sociology is not simply sociology of fisheries. While it can inform the study of fisheries, this approach draws out and elucidates the larger socio-ecological interactions and processes, i.e., developing an analysis of the interpenetration of human and marine systems.[7] Marine sociology moves beyond the important, but often non-ecological studies of fishing culture, industries, and policies. It submerges these sociological analyses into the ocean of ecological interactions. This approach opens up a vast realm of sociological investigation. It can result in increased potential for comprehension of socio-ecological concerns, as well as possible solutions to the ocean of troubles we must surely tackle in the coming decades. A marine sociology approach has been initiated in fisheries research, which includes drawing on the human ecology tradition (Hamilton et al. 2003; Hamilton et al. 2004) as well as social metabolism (Clausen and Clark 2005; Longo 2012; Longo et al. 2015). But much more sociological work needs to be done.

Sociology has theories and methods that can be employed to study human-marine relationships of the sort we suggest. We emphasize that social metabolic analysis is just one of the possibilities. Metabolic theorists recognize that ecosystems, including distinct natural cycles and process, operate independently of and in concert with human relationships. The universal metabolism of ecosystems allows for the conditions of nature that support human communities. It also marks natural limits, which when transgressed cause socio-ecological problems. These theorists specify that the social metabolism—which includes the interpenetration of humans and the rest of nature—is profoundly influenced by the prevailing political-economic arrangements. They suggest that the logic of capital creates a distinct social metabolic order, given its constant drive to expand and accumulate capital. This relentless growth imperative generates social tendencies, which violate the conditions that support the universal metabolism of nature and produce ecological rifts in natural cycles and processes. The approach clearly lends itself well to the socio-ecological study of marine systems.

Human ecology laid the groundwork for this research program, as it emphasized that including ecological conditions will further sociological analyses, and can integrate easily with social metabolic theory for developing marine sociology (Longo et al. 2015). Other theoretical approaches, like the treadmill of production, ecological modernization, and environmental justice raise important questions that can enhance this work as well. However, as these approaches are employed to marine sociology, we suggest that it will require an increasing treatment and incorporation of ecosystems in the studies. This is not implausible, as these are materialist theories, and this work has played an important part in expanding environmental sociology and the new ecological paradigm.

7 In a previous draft of this article, we included an analysis of overfishing employing the framework we set forth above. Nevertheless, following the suggestions of reviewers, we decided it would be best to leave it out. This decision also relates, in part, to the effort to present a perspective that can expand past fisheries and into other understudied marine realms.

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Making use of the analytical framing we have provided here, sociologists can continue to study the reciprocal interactions, feedback loops, and the indirect and direct relationships within and between social systems and marine systems. Furthermore, sociologists can better examine the social outcomes that result from ecological changes in marine systems, not in an isolated way, but in a manner that also directly refers back to their causes. These analyses can provide a sound basis for environmental policies that address the changes in marine systems and their social implications.

Developing what we call marine sociology points to the continuing need for increasing the ecological dimensions of environmental sociological investigations. Articulating this approach clarifies that some aspects of environmental research are still outside the mainstream of sociology as they are seen as independent of, or conflicting with, standard sociological research. Marine sociology is not a distinct subfield of sociology per se, but a program of study that furthers socio-ecological analyses, and in this sense, it is an important formulation for enhancing the power and quality of environmental sociological research, and sociology in general. Marine sociology underscores the need for a critical and integrative approach, which can help bridge the social/natural science divides and strengthen sociology's position in the policy debates associated with global environmental change.

In order to highlight the potential range of issues for marine sociology, we provided succinct analyses of climate change, ocean acidification, and pollution. Each discussion illuminates distinct anthropogenic forces that are part of the larger social metabolic order of capital and the types of ocean rifts that are being created in the marine systems. We only present these briefly, and each of these problem areas needs to be further investigated and elaborated using a sociological lens. Nevertheless, we give emphasis to the integration of sociology and ecology.

Our concise discussion demonstrates the promise of this approach, in particular the necessity to incorporate ecological processes and systems into social analysis to help clarify the broad range and complex developments that emerge. Marine sociology makes it apparent that this critical move is indispensable for environmental sociological research in the era of global environmental change. In addition, we suggest that there are numerous other marine concerns that are ripe for advancing socio-ecological analyses, including social problems associated with invasive species, oil and gas exploration, energy production (such as wind, wave, and tidal), biodiversity loss, aquaculture systems, coastal development, and specific forms of toxic pollution such as mercury, dioxin, and pesticides, among others. In some cases, useful work has begun. Yet, these concerns have been under-investigated by sociologists.

The links between terrestrial and marine activities as presented in this article should prompt us to consider whether these systems should be analytically separated at all. Marine sociology reveals how human and ocean systems are coupled, drawing attention to the interactions that shape life in both realms. We primarily focused on how humans are transforming marine systems and the various ecosystem consequences. Here we only touch on how these changes affect human societies. This part of the discussion also needs to be expanded in future work. Additionally, the various health implications of marine degradation need to be explored more extensively. Nevertheless, we aim to make these issues more visible for sociologists, while encouraging marine sociology as a legitimate realm of social inquiry.

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